



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A) A.Y. 2026 - 2027

TITLE OF THE PROJECT		
S. No.	University ID	Name
1	2420030298	Gowrie Reddy
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Abstract

CodeShield is an AI-driven vulnerability remediation pipeline designed to improve software security through automation. It uses Generative AI and Large Language Models (LLMs) to detect security vulnerabilities, analyse their causes, and suggest or generate suitable fixes. These fixes are automatically tested and validated through a DevSecOps-based CI/CD pipeline before deployment. By reducing manual effort and enabling faster vulnerability resolution, CodeShield helps developers build more secure and reliable software. The project demonstrates the use of Adaptive Software Engineering, Generative AI, automated testing, CI/CD, and DevSecOps in a practical software development environment.

CodeShield has a wide range of real-life applications across industries that handle sensitive data and critical software systems. In banking and financial services, it can automatically detect and help fix vulnerabilities in banking software, payment systems, and online transaction platforms, protecting sensitive financial information. In e-commerce, CodeShield can identify security issues such as SQL injection, insecure authentication, and exposed customer data before applications are deployed. Similarly, in healthcare, it can detect security weaknesses in hospital and healthcare applications to help protect sensitive patient and personal information. CodeShield can also be integrated into cloud-based applications and CI/CD pipelines on platforms such as AWS and Azure, enabling continuous vulnerability detection and remediation as software is developed and updated.

Signature of the Guide

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